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WEIGHT MACHINE SENSOR SYSTEM

Abstract

A load sensor includes a bolt configured to support a pulley of a weight machine, and a strain gauge laminated onto an exterior surface of a shoulder section of the bolt.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a divisional of U.S. application Ser. No. 18/051,824, filed Nov. 1, 2022, which claims priority to and the benefit of U.S. Provisional Application No. 63/263,463, filed Nov. 3, 2021, the entire disclosures of which are incorporated by reference herein.

BACKGROUND

[0002] The proliferation of fitness sensors has enabled more engaging user experiences for exercisers. Data produced from the sensors can be gathered and transmitted wirelessly in a variety of formats to enable more personalized exercise plans, more granular insights into human performance, and sharing of workout results across digital communities. Strength equipment, however, has lagged behind cardiovascular equipment due to unique challenges that prohibit efficient scaling.

TECHNICAL FIELD

[0003] The present disclosure relates to weight machines and, in particular, sensor systems therefor.

SUMMARY

[0004] Disclosed herein are implementations of weight machine sensor systems, sensors thereof, and methods related thereto. In one implementation, a load sensor includes a bolt configured to support a pulley of a weight machine, and a strain gauge laminated onto an exterior surface of a shoulder section of the bolt.

[0005] The shoulder section may include a stress concentration geometry that may axially overlap the strain gauge and may be a recess extending circumferentially at least partially around an axis of the bolt. The bolt may be configured to orient the strain gauge relative to the weight machine in a predetermined orientation, and may include an alignment feature that is a flat face of the shoulder section that mates with a corresponding alignment feature of a bracket of the weight machine. The corresponding alignment feature may be a non-circular through-hole. The predetermined orientation may include the strain gauge being in one of tension or shear when the pulley is loaded.

[0006] In an implementation, a weight machine sensor system is provided for determining a weight setting of a weight machine having weight plates and incremental weights to be lifted during an exercise. The system includes a load sensing bolt and a controller. The load sensing bolt is configured to support a pulley of the weight machine and includes a strain gauge for sensing force transferred from the weight plates and the incremental weights by the pulley to the load sensing bolt. The controller is in communication with the load sensing bolt, and determines a weight setting of the weight machine according to the load sensing bolt.

[0007] The weight sensor system may further include an incremental weight sensor configured to detect whether the force includes one or more incremental weights. The controller may be in communication with the incremental weight sensor and may determine the weight setting of the weight machine according to the load sensing bolt and the incremental weight sensor. The controller may calculate a weight measurement value from the force sensed by the load sensing bolt, and may determine the weight setting with a rounding algorithm according to the weight measurement value and a signal received from the incremental weight sensor indicating whether the incremental weights lifted during the exercise. The rounding algorithm may account for weight values, numbers, and sequencing of the weight plates.

[0008] In an implementation, a method is provided for retrofitting a weight machine with a weight machine sensor system that is configured to determine a weight setting of the weight machine during an exercise. The weight machine is of a type having a weight stack with weight plates that are to be lifted during the exercise. The method includes: providing a weight machine sensor system that includes a load sensing bolt and a controller; replacing a pulley bolt of the weight machine with the load sensing bolt to support a pulley of the weight machine, the load sensing bolt being configured to sense force applied by the pulley thereto; and calibrating the load sensing bolt to determine a weight function by which a weight measurement value can be determined from a

force value output by the load sensing bolt.

[0009] The weight machine may be of a type having incremental weights that are to be selectively lifted with the weight plates of the weight stack during the exercise. The weight machine sensor system may include an incremental weight sensor that may be configured to sense which of the incremental weights of the weight machine are being lifted or are to be lifted during the exercise. The method may further include installing the incremental weight sensor to the weight machine. The calibrating may include lifting the weight stack at different weight settings, measuring force values at the different weight settings, and determining the weight function according to the force values and the different weight settings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 shows the most common elements of a weight machines which delivers an exercise resistive force by raising a weight stack comprised of weight plates against the force of gravity using a cable or belt in tension to transfer that force.

[0011] FIGS. 2A to 2C illustrate some examples of cable linkage configurations that are used to raise and lower the weight stack of a weight machine.

[0012] FIGS. 3A and 3B provide views of two mechanisms for engaging incremental weights on a weight machine to provide finer adjustments to the weight setting than the primary weight plates in the weight stack allow.

[0013] FIGS. 4A to 4C illustrate existing hardware on a weight machine (FIGS. 4A and 4B) and a modification of the existing hardware to include load sensing bolt to extract force data for the purpose of enhanced digital experiences (FIG. 4C).

[0014] FIG. 5 illustrates a load sensing bolt in closer detail to show the functional components.

[0015] FIG. 6 shows a sample configuration of incremental weights that has integrated reed switch sensors to determine their usage state, thereby adding accuracy and longevity to the overall load sensing system.

[0016] FIG. 7 shows an alternative method for incremental weight usage detection that features reflective surfaces that vary in distance from a ranging sensor depending on the orientation of the selector knob.

[0017] FIGS. 8A and 8B show a calibration tool that can be used to mechanically raise and lower the weight stack of a weight machine to assist with the calibration process.

[0018] FIG. 9A is a schematic of a weight machine sensor system that may include the load sensing bolt and an incremental weight sensor.

[0019] FIG. 9B is a schematic of an example hardware configuration of a controller of the weight machine sensor system.

[0020] FIG. 10 is a flowchart of a method of retrofitting a weight machine with a weight machine sensor system.

[0021] FIG. 11 is a flowchart of a method of determining a weight machine setting during an exercise with a weight machine sensor system.

DETAILED DESCRIPTION

[0022] Exercise is critical for human physical and emotional health. Many people understand these benefits and strive to integrate exercise into their routines but fail because they become bored with the experience or do not get results. Digital user experiences for runners and cyclists, for example, have been popularized by using human performance data as a way to provide motivation, demonstrate progress, provide data-driven coaching instructions, and create content that can be shared in digital formats within online communities. Often human performance data can be captured through wearable technologies such as a smart watch, chest strap, foot pod, etc. In the

example of cycling there are well-established technologies such as cadence sensors and power meters that attach directly onto the bike. For resistance training, however, it is difficult to measure force in a consistent manner across the variety of machine configurations and therefore no such technologies have achieved meaningful market adoption.

[0023] Resistance training, also commonly called strength training or weightlifting, is one of the most common forms of exercise in the world and yet it has not experienced the same proliferation of digital user experiences due to the lack of appropriate sensor data. Heart rate monitors do not measure responses adequate for anaerobic activity and generally are more useful for measuring relative effort rather than actual performance. Furthermore, wearable technologies cannot capture human motion for extremities on which it is not attached. For example, a smart watch worn on the wrist is motionless while an exerciser is performing a seated leg curl. Most importantly, the key metric for resistance training is the amount of resistance used for that exercise—typically measured in force and/or the mass moved against gravity—and such metrics are difficult to measure without sensors directly integrated into the source of resistance itself. The weight machine force sensor described herein produces this resistance data from conventional weight machines that are widespread within the fitness industry yet have remained mostly non-digitized to date.

[0024] A common form of resistance training uses a machine designed to mechanically isolate exerciser motion to target the desired muscle group. Motion resistance is usually provided through cable tension directed through a series of pulleys and cams to a resistance source. The difficulty level depends on user selected resistance based on ability and training goals. One form of resistance machine, commonly called “selectorized” features a stack of weights configured such that the exerciser selects a quantity of weight by inserting a pin, flipping a paddle, or any other such mechanism for isolating which weight plates are moved. These weight plates could be in fixed increments of 10, 15, or 20 lbs, for example, to make the values easier for the exerciser to remember. Some machines have consistent weight plate values while others have weight stacks that feature two or more major weight plate values. For example, the weight stack could use a stack of twenty 15 lb plates for a total mass of 300 lbs or it could feature five 20 lb plates at the bottom of the stack then twenty 10 lb plates at the top of the stack to achieve the same total mass of 300 lbs. During the exercise motion the weights selected by the exerciser are typically moved upward against the direction of gravity, so the level of resistance is proportional to the quantity of weight selected. In some cases, however, selectorized machines can be designed such that the weight selected reduces the amount of resistance by effectively subtracting that resistance from the exerciser's body weight during body weight exercises. In these scenarios the weight moves downward with the direction of gravity during the exercise motion. This is applicable for pull ups or assisted chin dips, for example, where the exerciser's body weight is the source of resistance.

[0025] FIG. 1 shows a sample weight machine 200 resistance mechanism in which the weight stack 202 is comprised of weight plates 204 of uniform mass. A cable 208 or belt is used to transfer a resistive force imparted by the exerciser during the exercise motion to the weight stack 202 causing it to move against the force of gravity 206. Guide rods 210 are used to constrain the motion of the weight stack 202 to the intended vertical direction. Any configuration of levers, cams, pulleys 212, etc. can be used to redirect the exercise resistive force 214 delivered by the exerciser to the weight stack 202. The weight machine 200 can take on a wide variety of possible mechanical configurations depending on the exercise motion and its target muscle groups, but they commonly feature at least one pulley 212 directly above the weight stack 202 to ultimately redirect the cable 208 down toward the weight stack 202.

[0026] FIGS. 2A to 2C show three examples of the final cable 208 routing patterns that could be used to lift the weight stack 202 via the exercise resistive force 214 delivered from the exerciser. In FIG. 2A the cable 208 terminates directly into the weight selection bayonet 216 that is used to isolate which number of weight plates 204 will move during the exercise motion. In FIG. 2B the weight selection bayonet 216 is connected via a linkage to a weight stack pulley 220 that receives a

cable **208** transmitting the exercise resistive force **214**. In this example the cable **208** on one side of the weight stack pulley **220** terminates at a fixed anchor **222** while the other side is connected to move via the exercise resistive force **214** delivered during the exercise motion. FIG. 2C shows a similar configuration but rather than one side of the cable **208** terminating in fixed anchor **222**, the cable **208** on both sides of the weight stack pulley **212** are redirected via pulleys **212** to transfer exercise resistive force **214** delivered during the exercise motion. Note that in each of these examples a stationary pulley **212** is used to redirect the cable **208** which is in tension due to the exercise resistive force **214** it is redirecting.

[0027] Some weight machines **200** will also allow the exerciser to make finer resistance setting adjustments by allowing additional incremental weights **224** to be combined with the major weight plates **204**. If the major weight plate values are 15 lbs then the exerciser could be enabled to optionally add 5 or 10 lbs to achieve more granular control over the amount of weight lifted. For example, if the major weight plates **204** are 15 lbs and the exerciser had been lifting 100 lbs the next highest setting would be 115 lb if no incremental weights **224** were available, but by using the incremental weights **224** with incremental weight values of 5 lbs the intermediate values of 105 or 110 lbs can also be selected by adding one or two 5 lb incremental weights **224**, respectively.

[0028] FIGS. 3A and 3B show a sample configuration for a weight stack **202** featuring incremental weights **224**. The incremental weights **224** are typically engaged by the exerciser at the same time he/she selects the major weight plate values. When selected, the increment weights are functionally connected to the primary weight stack **202** and move in unison. The incremental weights **224** can be positioned behind the weight stack **202** and engaged or disengaged by rotating a selector knob **226** as shown in FIG. 3A or with a push/pull slider **228** as shown in FIG. 3B. As these two examples show there can be a single incremental weight **224** or two or more incremental weights **224**. Alternatively, the incremental weights **224** could be mounted above the weight stack **202** then lowered such that they rest on top of the weights stack for additional mass when increased resistance is desired. In another embodiment the incremental weights **224** could be slid from a stationary storage position adjacent to the primary weight stack **202** onto the weight stack **202** or manually lifted and set on top of the weight stack **202** when desired.

[0029] An important metric in resistance training is the amount of resistance exerted by the exerciser during the exercise motion. The mass that is moved against gravity, or the weight setting, is an important factor in determining the resistance but the amount of acceleration imparted on that mass by the exerciser also contributes. The term weight setting generally refers to an amount of weight (e.g., a number of pounds) that corresponds to the combination of the weight plates **204** of the weight stack **202** and the incremental weights **224** that are lifted during an exercise. The resistive force is proportional to not only the mass but also its acceleration and deceleration relative to gravity. During a repetition with a 100 lb weight setting, the cable **208** will transmit resistive force values significantly in excess of 100 lbs-f while accelerating or significantly below 100 lbs-f while decelerating and only achieves a stable 100 lbs-f if the weight stack **202** is held static. The force data therefore provides much more granular insight into human performance including maximum force, peak force per repetition, rate of force development, fatigue, etc.

[0030] One method for force measurement on weight stack **202** machines involves a device that is attached to the cable **208** above the weight stack **202**, but on some weight machines **200** this can restrict the range of motion because the physical footprint of the sensor restrains the motion the weight stack **202** can travel at its topmost extent. Another method is to configure an additional pulley **212** connected with a force transducer into the weight machine **200**. If the pulley **212** changes the direction of the cable **208** then it can be used to detect the tension, or resistance transferred through that cable **208**. This approach adds moving parts and therefore complexity, however. The additional cable **208** re-directions can also affect the life of the cable **208** on those machines which is impacted by the number of times it is flexed during routine operation.

[0031] Capturing force data while minimizing impact on the complexity, range of motion, or life of

a selectorized weight machine **200** is beneficial. One way to do so is to measure the force transmitted through one of the pulleys **212** already present in the machine's design. Minimizing the changes to the design not only preserves simplicity but also enables the millions of machines already in existence to become retrofit with a sensor and therefore enabled to be capable of facilitating more engaging, digital user experiences.

[0032] FIG. **4A** shows a more detailed view of a pulley **212** on a weight machine **200** which in this example is positioned directly above the weight stack **202**. A structural bracket **230** fixed to the weight machine **200** contains two vertical sidewalls **232** with bolt through-holes **234** designed to receive a pulley bolt **236**. The pulley bolt **236** extends through both vertical sidewalls **232** of the structural bracket **230** and through the pulley bearing **238** such that the pulley **212** is positioned between the two vertical sidewalls **232**. The pulley bolt **236**, thereby, supports the pulley **212** relative to the weight machine **200** and, in particular, relative to the structural bracket **230**. In another example, there could be a single sidewall with a pulley **212** mounted using a pulley bolt such that the pulley **212** is cantilevered off the side. The pulley bolt is secured using a washer **240** and nut **242**. The vibrations experienced by the machine during routine use could cause the nut **242** to loosen, so a locking nut **242**, locking washer **240**, or locking compound on the threads of the pulley bolt could be used to ensure the nut **242** remains in position. In this example there is also a cable retention bracket **244** which is used to ensure that the cable **208** remains routed properly over the pulley **212**. FIG. **4B** shows an exploded view of the same configuration shown in FIG. **4A** to show more detail.

[0033] FIG. **4C** shows a revised configuration in which the pulley bolt **236** has been replaced by a load sensing bolt **246**. In this example the load sensing bolt **246** is sized to directly replace the pulley bolt **236** show in FIGS. **4A** and **4B** and functions to support the pulley **212** relative to the structural bracket **230** and sense the force applied by the pulley **212** thereto. The load sensing bolt **246** features a head **248** on one end followed by a smooth, unthreaded shoulder section **250**. The diameter of the bolt **246** is then reduced for the final threaded section **252** which is capable of receiving a fastener such as a nut **242**. The shoulder section **250** and the final threaded section **252** may be considered to form a shaft of the load sensing bolt **246**, and with the head **248** may be considered to form a bolt.

[0034] The load sensing bolt **246** is configured to rotationally align with the weight machine **200** (e.g., the structural bracket **230**), so as to orient strain gauges **260** (discussed in further detail below) relative to the weight machine **200** and/or relative to the force applied by the pulley **212** from the cable **208** (i.e., when the exercise resistive force **214** is applied thereto by the exerciser) in a predetermined orientation. The shoulder section **250** of the load sensing bolt **246** in this example features one or more (e.g., two) flat faces. A first of the flat faces may be an orientation face **254** that serves as the alignment feature to ensure that the load sensing bolt **246** is installed at the appropriate angle (i.e., the predetermined orientation) to detect force transmission with a high degree of resolution. The pulley bracket through-hole **256** in FIG. **4C** is not circular, but rather specifically shaped to aid in the proper orientation of the load sensing bolt **246**, and thereby functions as a complementary alignment feature to that of the orientation face **254** of the load sensing bolt **246**. In this example the orientation face **254** is planar and will therefore align with flat section of the pulley bracket through-hole **256**. The pulley bracket through-hole **256** may be part of the structural bracket **230**, as originally provided or as modified to include. Alternatively, the pulley bracket through-hole **256** may be part of another bracket, such as a replacement to the cable retention bracket **244**, the existing cable retention bracket **244** with a modification, or another bracket, that couples to the structural bracket **230** in another predetermined orientation. The pulley bracket through-hole **256**, thus, orients the load sensing bolt **246** to the bracket and, thereby, to the structural bracket **230** and the weight machine **200**.

[0035] A second of the flat faces of the shoulder section **250** may be a strain gauge face **258** that is designed to provide a surface for laminating one or more strain gauges **260** to the load sensing bolt

246. In one example of the predetermined orientation, the load sensing bolt **246** is oriented for the strain gauge face **258** and, thereby, the strain gauge **260** to be in tension when the resistive force **214** is applied to the cable **208** (e.g., generally or substantially opposite to a net force applied by the pulley **212** to the load sensing bolt **246**). In another example of the predetermined orientation, the load sensing bolt **246** is orientated for the strain gauge face **258** and, thereby, the strain gauge **260** to be in shear when the resistive force **214** is applied to the cable **208** (e.g., substantially parallel to a net force applied by the pulley **212** to the load sensing bolt **246**).

[0036] FIG. 5 shows the load sensing bolt **246** with the strain gauge face **258** oriented to display a sample orientation of strain gauges **260**. One or more strain gauges **260** could be laminated onto a single surface, as shown, or on multiple strain gauge faces **258**. The strain gauges **260** are resistive elements configured such that their resistance changes when the material to which they are laminated becomes deflected under load. The change in resistance is proportional to the amount of deflection caused by the load. The strain gauges **260** can be configured in a half-bridge or full-bridge configuration of a Wheatstone Bridge circuit such that the differential voltage signal indicates the magnitude of the load applied. An amplifier and processor on the weight machine **200** are used to interpret this data for calculation of exercise metrics. The output of the load sensing bolt **246** and, specifically, of the one or more strain gauges **260** (e.g., the differential voltage value) may be referred to as a force signal, which includes a force value (e.g., the voltage) that is indicative of the force sensed thereby.

[0037] Stress concentration geometries **262** such as the ones shown can be used to isolate the deflection into the region on which the strain gauges **260** are applied for an improved load measurement signal. Each stress concentration geometry **262** is a feature that concentrates bending stress to a region proximate one of more of the strain gauges **260**. For example, as shown, the stress concentration geometry **262** may be a recess that protrudes radially into the load sensing bolt **246** (e.g., the shoulder section **250**) and that is axially overlapping one or more of the strain gauges **260** (i.e., sharing a common location along an axis of the bolt **246**) or axially adjacent to one or more of the strain gauges **260** (e.g., being within 10 mm, 5 mm, 2 mm, or less). The recess of the stress concentration geometry **262** may, for example, extend circumferentially at least partially the axis of the bolt **246** (e.g., other than the flat faces), be positioned generally radially opposite one or more of the strain gauges **260**, have a uniform depth, and/or have a uniform cross-sectional shape (e.g., being partially circular).

[0038] In the example shown the strain gauge wires **264** enter a hole **266** in the orientation face exit via an axial cavity **268** in the load sensing bolt **246** then out the head **248**. The head **248** of the load sensing bolt **246** could also feature a recessed cavity that could be filled with an epoxy or potting compound **270** to protect the wires exiting the load sensing bolt **246** and prevent them from being inadvertently damaged during installation or servicing. The length of the shoulder could be sized such that when the nut **242** is fully torqued onto the threads the vertical sidewalls **232** of the structural bracket **230** are compressed to a position that constrain the pulley **212** to a limited amount of lateral movement without allowing any portion of the pulley **212** to be clamped by the vertical sidewalls **232** which would impair force measurement.

[0039] The load sensing bolt **246** may be used when the structural bracket **230** is already designed or already deployed on weight machines **200** in the field. The diameter of the shoulder, for example, may be limited to the size of the bolt through-holes **234** of the vertical sidewalls **232** through which it must fit. In some scenarios the bolt through-holes **234** could be bored out larger to accommodate a larger diameter load sensing bolt **246** but this step could be skipped to eliminate the need for power tools for installation on machines which have already been field-deployed. The amount of surface area available for laminating the strain gauges **260** is therefore limited and the mechanical fixturing of the load sensing bolt **246** is limited to the structural elements originally designed for standard bolts.

[0040] Additional sensor inputs on the weight machine **200** can be used to supplement the data

from the load sensing bolt **246** to preserve accuracy of the exercise metrics over the lifetime of the weight machine **200**. For example, a processor on the machine **200** (e.g., the controller **296**, as discussed below) can use the data produced by the load sensing bolt **246** to determine the weight setting used during an exercise but in some scenarios, this could require a high level of precision. For example, if the weight of the incremental weight **224** is 5 lbs then detecting the accuracy within the maximum deviation of 2.5 lbs on a weight stack **202** of 400 lbs would require an error of no more than 0.6%. If an additional sensor, such as an incremental weight sensor **297** (discussed below with respect to FIG. **9A**), could detect the usage state of the incremental weights **224**, however, then the load sensing bolt **246** would only need enough accuracy to detect error bounds dictated by the mass of the weight plates **204**. So if the weight machine **200** had 15 lb weight plates **204** then the maximum deviation of 7.5 lb on a weight stack **202** of 400 lbs would require an error of no more than 1.9%, which is a more reasonable performance target particularly for space-constrained low cost force measurement technologies.

[0041] The incremental weight sensor **297** may be configured to detect which of the incremental weights **224** are being lifted or are to be lifted with the weight stack **202** during an exercise, for example, by detecting movement of the one or more incremental weights **224** (e.g., with reed switches or other magnetic sensors, physical contact switches, and/or optical sensors) and/or by detecting a setting indicative of which of the incremental weights **224** are to be lifted (e.g., optically). For example, the incremental weight sensor **297** may be configured to detect the movement or position of a detection component **298** that may be attached to an incremental weight selection device (e.g., the selector knob **226**) or to the incremental weights **224** themselves. An output of the incremental weight sensor **297** may be referred to as an incremental weight signal, which is indicative of the incremental weight value being lifted or to be lifted for a given incremental setting. Various configurations of the incremental weight sensor **297** are discussed in further detail below with respect to FIGS. **6** and **7**.

[0042] FIG. **6** shows a view of the rear of the weight stack **202** featuring two incremental weights **224** which are engaged or disengaged via a selector knob **226** by the exerciser. In this example the exerciser can rotate the knob **226** to select one incremental weight **224** to add 5 lbs of weight or two incremental weights **224** to add 10 lbs. As noted previously there are many possible mechanisms for adding and subtracting incremental weights **224** and FIG. **6** is illustrating just one example. In this example the incremental weights **224** have magnets **272** attached which are detected by reed switches **274**, which may be or otherwise form the incremental weight sensor **297** and the detection component **298**, respectively, when in their resting position on the pedestal **276**. If the increment weights are selected then they are joined with the weight stack **202** and therefore in motion during the exercise. The reed switches **274** will detect an absence of the incremental weights **224** because the magnetic field exceeds their detection range when the weights are lifted. The reed switches **274** could also be positioned such that they are proximal to the magnets **272** mounted on the incremental weights **224** but then detect the engagement of the increment weights as the magnets **272** move past the reed switch during the exercise motion when the weight stack **202** is raised and lowered. Reed switches **274** are just one example of a type of sensor that can be used for detecting incremental weight usage, which may form the incremental weight sensor **297**. In a further example, the incremental weight sensor **297** may include or be formed by one or more physical contact switches which are mechanically depressed could be used or optical sensors directed at the incremental weights **224** to determine their presence or absence during the exercise motion. In a still further example, the incremental weight sensor **297** may include or be formed by one or more reed switches **274** available from manufacturers like Magnasphere Corporation that also provide a magnetic field for detection of ferrous objects (e.g., the incremental weights **224**, themselves) could also be used. This still further approach would eliminate the need for attaching magnets **272** to the incremental weights **224**. The reed switches **274** output the incremental weight signal, which indicates which of the incremental weights **224** were lifted and/or the incremental

weight value corresponding thereto.

[0043] FIG. 7 demonstrates another approach for detecting the usage of incremental weights 224 by using a ranging sensor 278, which may be or otherwise form the incremental weight sensor 297. These sensors use technologies such as light detection and ranging (LIDAR), ultrasonic ranging, or infrared ranging to detect the distance of objects toward which they are oriented. If the user selects the incremental weight 224 by rotating a knob 226 then multiple reflective surfaces 280, which may form the detection component 298, could be integrated into that knob 226 such that the reflective surface 280 is a different distance from the ranging sensor 278 depending on the orientation of the knob 226. A push-pull mechanism or other mechanisms could similarly be used to position a reflective surface 280 with a different distance from the ranging sensor 278 depending on the selection made by the exerciser. Multiple ranging sensors 278 could also be used such that one measures the distance to a fixed surface, the other measures the distance to a surface that varies depending on the incremental weight 224 selection, and the difference between the two values is used to determine the selection status of the incremental weights 224. The ranging sensor 278 outputs the incremental weight signal, which indicates which of the incremental weights 224 are to be lifted or the incremental weight value corresponding thereto.

[0044] As mentioned, a benefit of using a load sensing bolt 246 to replace existing bolts is that it can be used to retrofit weight machines 200 already in use in the field to enable enhanced, data driven user experiences. To function properly, however, the load sensing bolt 246 should be calibrated after it has been installed onto the weight machine 200. One way to perform a calibration is to lift the weight stack 202 then measure the signal coming from the load sensing bolt 246 at multiple different weight settings. The relationship between the weights being lifted and the signal received from the load sensing bolt 246 define a calibration fit (e.g., a weight function) which is then stored on the weight machine's 200 processor to determine exercise metrics in real-time. Lifting a large fraction of the weight stack 202 is often necessary to perform a calibration, but technicians and operators in the field are often incapable of performing such a task due to physical or regulatory limitations. Therefore a device to mechanically assist in the calibration procedure would be another important component of the concept described herein.

[0045] FIGS. 8A and 8B show a calibration tool 282 for lifting the weight stack 204 during calibration. The calibration tool 282 includes a rigid frame 284 with three pulleys 286, 288, two of which are fixed pulleys 286 attached to the frame 284 and a third movable pulley 288 which is actuated along a track with a threaded rod 290 connected to the rigid frame 284. The threaded rod 290 is configured to provide sufficient mechanical to be driven by a drive source (e.g., an electric motor or drill as discussed below) and to prevent back driving from the weight stack 204. The movable pulley 288 could also be attached via a hinged connection or 4-bar mechanism to provide the actuated motion relative to the other fixed pulleys 286. FIG. 8A shows the calibration tool 282 in its initial state before the weight stack 202 has been lifted and FIG. 8B shows the tool in its secondary state after the movable pulley 288 has been actuated which causes the cable 208 length to be effectively shortened thereby lifting the weight stack 202. The length of cable 208 available to the calibration tool 282 varies from machine to machine and in some cases there is limited space above the weight stack 202. Not all of the pulleys 212 need to be able to rotate for the mechanism to work so they could be trimmed down and fixed to the rigid frame 284 or replaced with cams to help reduce the height of the tool. The example shown features a spindle 292 at the end of the threaded rod 290 which engages with the chuck of a handheld drill 294, but the calibration tool 282 could also include its own electric motor and drivetrain to eliminate the need for the handheld drill 294 or other external actuator.

[0046] FIG. 9A is a schematic illustration of a weight machine sensor system 295 that generally includes a controller 296 and the load sensing bolt 246 and may further include the incremental weight sensor 297. The controller 296 is in communication with the load sensing bolt 246 and the incremental weight sensor 297, if provided, and determines a weight setting according to the force

sensed by the load sensing bolt **246** and the incremental weight signal received from the incremental weight sensor **297** (if included). The controller **296** may further be in communication with an external device **299**, directly or indirectly, such as a personal electronic device of the exerciser (e.g., a smart watch or smartphone) or a computing system (e.g., a local and/or remotely-located computing system) to which the controller **296** may send weight information (e.g., the weight setting, repetitions, and timing, among other information).

[0047] FIG. **9B** illustrates one example hardware configuration for a controller **296**, though it should be understood that any device capable of performing the functions and instructions of the controller **296** may be utilized. The controller **296** generally includes a processor **296a**, a storage **296b**, a memory **296c**, a communications interface **296d**, and a bus **296e** by which the other components or aspects of the controller **296** are in. The processor **296a** may be any suitable processing device, such as a central processing unit (CPU), capable of executing written instructions (e.g., software) to perform the methods described herein. The storage **296b** may be any long term, non-volatile memory device, such as a solid state drive capable of storing the instructions executed by the processor **296a**. The storage **296b** may be considered to be or include a computer readable medium. The memory **296c** is a short term, volatile memory device, such as a random access memory module. The communications interface **296d** may be any device or combination of devices configured to send and/or receive signals (e.g., to and/or from the load sensing bolt **246** and the incremental weight sensor **297**, and/or to the external device **299**).

[0048] Referring to FIG. **10**, a weight machine **200** may be retrofit with the weight machine sensor system **295**, which is then calibrated to that particular weight machine **200**.

[0049] To retrofit the weight machine **200** with the weight machine sensor system **295**, the pulley bolt **236** is replaced with the load sensing bolt **246** (e.g., the pulley bolt **236** is removed, and the load sensing bolt **246** is inserted through the structural bracket **230** and the pulley **212** to support and couple the pulley **212** to the structural bracket **230**). The load sensing bolt **246** is also oriented relative to the weight machine **200** in the predetermined orientation. The weight machine **200** is also retrofit to include the alignment feature (e.g., a flat face) that corresponds to the orientation face **254** of the load sensing bolt **246**, for example, by retrofitting the structural bracket **230** to include the alignment feature (i.e., the pulley bracket through-hole **256** that includes a flat face or is otherwise not circular), retrofitting the cable retention bracket **244** to include the alignment feature, or replacing the cable retention bracket **244** with another cable retention bracket **244** that includes the alignment feature and itself orients to the weight machine **200** (e.g., the structural bracket **230**) in another predetermined orientation.

[0050] To calibrate the weight machine sensor system **295**, the weight machine **200** is placed into different weight settings, the weight stack **202** is lifted (e.g. with the calibration tool **282**, as described above), force values (e.g., voltages) are recorded for each of the different weight settings, and a weight function is fit to the force values at the different weight settings (e.g., a second order polynomial). The different weight settings may, for example, include at least three different weight settings, one at a lower weight setting (e.g., lifting relatively little weight, such as one or two of the weight plates **204**), one at a higher weight setting (e.g., lifting relatively high weight, such as all but one or two of the weight plates **204**), and one or more at a middle weight setting (e.g., lifting weight approximately midway between the lower and higher weight settings). During use, the weight function is used in conjunction with force values from the force signal to determine a measured weight value and also determine the weight setting, as described in further detail below with respect to FIG. **11**.

[0051] The controller **296** is further programmed, either during the calibration process or a separate step, with characteristics of the weight machine **200** and, in particular, weight values of the weight plates **204** of the weight stack **202** (e.g., the weight plate values) and the incremental weights **224** (e.g., the incremental weight values. This programming may further include other information, such as the number of the weight plates **204** at each of one or weight plate values, sequencing of

the weight plates **204** at each of the weight plate values (e.g., two 30 lb weight plates are required before adding any of ten 15 lb weight plates), and the number of the incremental weights **224** at each of the one or more incremental weight values.

[0052] FIG. **10** is a flowchart of a method **1000** for retrofitting a weight machine **200** with the weight machine sensor system **295**. The method **1000** generally includes providing **1010** the weight machine sensor system **295**, installing **1020** the load sensing bolt **246** of the sensor system **295**, inputting **1030** weight machine information to the weight machine sensor system **295**, and calibrating **1040** the load sensing bolt **246**. The method **1000** may further include installing **1050** the incremental weight sensor **297** of the weight machine sensor system **295**.

[0053] The providing **1010** of the weight machine sensor system **295** generally includes providing the load sensing bolt **246** and the controller **296** and may further include providing the incremental weight sensor **297** (if included).

[0054] The installing **1020** of the load sensing bolt **246** generally includes replacing an existing pulley bolt **236** of the weight machine **200** with the load sensing bolt **246** (e.g., removing the existing pulley bolt **236** and inserting the load sensing bolt **246** to support the pulley **212** relative to the weight machine **200**). The installing **1020** of the load sensing bolt **246** may further include retrofitting an alignment feature to the weight machine **200** (e.g., retrofitting the pulley bracket through-hole **256** to be non-circular in the existing structural bracket **230** or the existing cable retention bracket **244**, or providing a new cable retention bracket **244** or another bracket that includes the pulley bracket through-hole **256** that is non-circular), and may still further include orientating the load sensing bolt **246** relative to the weight machine **200** in the predetermined orientation (as described previously). The installing **1020** may further include connecting the load sensing bolt **246** with the controller **296**, for example, with the gauge wire **264** to transfer power and signals therebetween.

[0055] The inputting **1030** of the weight machine information generally includes providing information about the weight plates **204** of the weight stack **202** and the incremental weights **224** into the controller **296** with any suitable device (e.g., a computing device connectable to the controller **296**). The information may, for example, include the weight values, numbers, and sequencing of the weight plates **204** of the weight stack **202** and the incremental weights **224**.

[0056] The calibrating **1040** of the load sensing bolt **246** generally includes lifting **1040-1** the weight stack **202** repeatedly at one or more different weight settings, measuring **1040-2** a force value at each of the different weight settings, and determining **1040-3** a weight function for the force values at each of the different weight settings.

[0057] The lifting **1040-1** of the weight stack **202** repeatedly at the different weight settings may be performed with the calibration tool **282** as described above or any other suitable device or method. The different weight settings may, for example, include relatively low, relatively high, and middle weight settings, as described previously.

[0058] The sensing **1040-2** of the force value at each of the different weight settings includes sensing a force value with the load sensing bolt **246**, and sending the force value in a force signal from the load sensing bolt **246** to the controller **296**. The force value may, for example, be a voltage output by the load sensing bolt **246**.

[0059] The determining **1040-3** of the weight function may be performed by the controller **296** in any suitable manner, for example, by filtering the force signals to determine singular force values at each of the weight settings and using a curve fitting algorithm to determine the weight function from the singular force values and the weight settings. As referenced above, the weight function may be a second order polynomial but may be any other suitable function appropriate for the characteristics of the strain gauges **260** (e.g., linear function). During subsequent use, force values (e.g., voltages) may be input to the weight function, which then outputs a weight measurement value.

[0060] The installing **1050** of the incremental weight sensor **297** generally includes attaching the

incremental weight sensor **297** to the weight machine **200**, as may be required by the type of the incremental weight sensor **297**, and attaching any corresponding detection component **298** (e.g., the magnets **272** or the reflective surface **280**) to the weight machine **200** (e.g., to the incremental weights **224** or the incremental weight selection device). The installing **1050** of the incremental weight sensor may further include connecting the incremental weight sensor **297** with the controller **296**, for example, via a wired connection to transfer power and signals therebetween.

[0061] Variations of the method **1000** are contemplated. For example, the load sensing bolt **246** may be initially installed in a new weight machine **200** (i.e., not retrofitting) and/or the calibrating **1040** may be performed as separate method.

[0062] Referring to FIG. **11**, a weight setting of the weight machine **200** may be determined according to the load sensing bolt **246** alone or in conjunction with the incremental weight sensor **297**. The load sensing bolt **246** sends the force signal to the controller **296**, which includes the force value (e.g., the voltage) indicative of the force applied to the load sensing bolt **246**. The incremental weight sensor **297** sends the incremental weight signal to the controller **296**, which includes information indicative of the incremental weights **224** being lifted or to be lifted (for a given incremental weight setting).

[0063] To determine the weight setting, the controller **296** receives the force value via the force signal from the load sensing bolt **246**, and inputs the force value into the weight function (determined from the calibration process, described previously) to obtain a measured weight value. The measured weight value is then rounded, using any suitable rounding function or algorithm, to the nearest possible weight setting determined according to the weight values, numbers, and sequencing of the weight plates **204** and the incremental weights **224**, as well as the incremental weight signal.

[0064] In a first example, the rounding algorithm rounds the weight measurement value to the nearest increment of the smallest incremental weight value to determine the weight setting. For example, if the weight measurement value is 182 lbs and the smallest incremental weight value is 5 lbs, the weight setting would be determined to be 180 lbs. This first example of the rounding algorithm may be performed without the incremental weight signal (e.g., in embodiments of the weight machine sensor system **295** without the incremental weight sensor **296**).

[0065] In a second example, the rounding algorithm first subtracts the known incremental weight values (i.e., from the incremental weight signal) from the measured weight value, then subtracts the plate weight values until achieving a negative value, and then compares the magnitudes of the negative value and the last positive value with the lower magnitude corresponding to the weight setting. By first subtracting the known incremental weight values and subsequently the plate weight values, the rounding algorithm allows for greater error in the weight measurement value while still determining the correct weight setting. For example, Table 1 represents this rounding algorithm applied to a weight measurement of 182 lbs on a weight machine **200** having weight plate values sequenced as 30 lbs for plates 1 and 2, 15 lbs for plates 3 to 10, and incremental weight values of 5 lbs for two incremental weights of which one is known to be used (i.e., via the incremental weight signal).

TABLE-US-00001

TABLE 1 Decremental Incremental/ Running Weight Plate Weight Weight Measurement Value Setting Initial Weight 182 Measurement Incremental 177 5 5 Weight Plate 1									
147	30	35	Plate 2	117	30	65	Plate 3	102	15
80	Plate 4	87	15	95	Plate 5	72	15	110	Plate 6
57	15	125	Plate 7	42	15	140	Plate 8	27	15
155	Plate 9	12	15	170	Plate 10	-3	15	185	

[0066] At plate 10, the decremented weight measurement is negative at -3 lbs. At plate 9, the last positive decremented weight measurement is 12 lbs, which has a magnitude that is greater than that of -3 lbs at plate 10. Therefore, the weight setting is determined to be that corresponding to plate 10, which is 185 lbs.

[0067] FIG. **11** is a flowchart of a method **1100** for determining a weight setting of a weight machine with the weight machine sensor system **295**. The method **1100** generally includes sensing

1110 force, sending **1120** a force signal, sensing **1130** the incremental weights with the incremental sensor **297**, sending **1140** an incremental weight signal from the incremental sensor **297** to the controller **296**, and determining **1150** the weight setting with a rounding algorithm. The method **1100** may further include sending a weight setting signal to an external device (e.g., a computing device of the exerciser or a computing system).

[0068] The sensing **1110** of the force and the sending **1120** of the force signal are performed with the load sensing bolt **246**.

[0069] The sensing **1130** of the incremental weights and the sending **1140** of the incremental weight signal are performed with the incremental weight sensor **297**, and may be performed in parallel with the sensing **1110** and the sending **1120**. The sensing **1130** and the sending **1140** are not performed by embodiments of the weight machine sensor system **200** that do not include the incremental weight sensor **297**.

[0070] The determining **1150** of the weight setting is performed with the controller **296**. The determining **1150** includes calculating **1150-1** a weight measurement from the force value (e.g., the voltage) using the weight function (e.g., as determined during the calibration process described previously). The rounding **1150-2** of the weight measurement to the nearest possible weight setting according to any suitable algorithm, including those describe previously (e.g., rounding the weight measurement value to the nearest value at the lowest value of the incremental weights, or subtracting the known incremental weights and weight plate values until achieving a negative value, the weight setting corresponding to the lower magnitude of the negative value or the lowest positive value).

[0071] The sending **1160** of the weight setting signal is performed with the controller **296**. The weight setting signal includes information indicating the weight setting for a particular exercise, and is sent to one of the external devices **299** (e.g., of the exerciser or a computing system).

[0072] In general, the computing systems and/or devices described may employ any of a number of computer operating systems, including, but by no means limited to, versions and/or varieties of the Microsoft Windows® operating system, the Unix operating system (e.g., the Solaris® operating system distributed by Oracle Corporation of Redwood Shores, California), the AIX UNIX operating system distributed by International Business Machines of Armonk, New York, the Linux operating system, the OS X, macOS, and iOS operating systems distributed by Apple Inc. of Cupertino, California, and the Android operating system developed by Google, Inc. and the Open Handset Alliance. Examples of computing devices include, without limitation, a computer workstation, a server, a desktop, notebook, laptop, or handheld computer, or some other computing system and/or device.

[0073] Computing devices generally include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above. Computer-executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Visual Basic, Java Script, Perl, etc. Some of these applications may be compiled and executed on a virtual machine, such as the Java Virtual Machine, the Dalvik virtual machine, or the like. In general, a processor (e.g., a microprocessor) receives instructions, e.g., from a memory, a computer-readable medium, etc., and executes these instructions, thereby performing one or more processes, including one or more of the processes described herein. Such instructions and other data may be stored and transmitted using a variety of computer-readable media.

[0074] A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Non-volatile media may include, for example, optical or magnetic disks and other persistent memory. Volatile media may

include, for example, dynamic random access memory (DRAM), which typically constitutes a main memory. Such instructions may be transmitted by one or more transmission media, including coaxial cables, copper wire and fiber optics, including the wires that comprise a system bus coupled to a processor of a computer. Common forms of computer-readable media include, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, a CD-ROM, DVD, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, a RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer can read.

[0075] Databases, data repositories or other data stores described herein may include various kinds of mechanisms for storing, accessing, and retrieving various kinds of data, including a hierarchical database, a set of files in a file system, an application database in a proprietary format, a relational database management system (RDBMS), etc. Each such data store is generally included within a computing device employing a computer operating system such as one of those mentioned above, and are accessed via a network in any one or more of a variety of manners. A file system may be accessible from a computer operating system, and may include files stored in various formats. An RDBMS generally employs the Structured Query Language (SQL) in addition to a language for creating, storing, editing, and executing stored procedures, such as the PL/SQL language mentioned above.

[0076] In some examples, system elements may be implemented as computer-readable instructions (e.g., software) on one or more computing devices (e.g., servers, personal computers, etc.), stored on computer readable media associated therewith (e.g., disks, memories, etc.). A computer program product may comprise such instructions stored on computer readable media for carrying out the functions described herein.

[0077] With regard to the processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. In other words, the descriptions of processes herein are provided for the purpose of illustrating certain embodiments, and should in no way be construed so as to limit the claims.

[0078] Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. In sum, it should be understood that the application is capable of modification and variation.

[0079] All terms used in the claims are intended to be given their ordinary meanings as understood by those knowledgeable in the technologies described herein unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary.

[0080] The Abstract is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the

following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A load sensor comprising: a bolt configured to support a pulley of a weight machine; and a strain gauge laminated onto an exterior surface of a shoulder section of the bolt for sensing force applied by the pulley to the bolt.
 2. The load sensor of claim 1, wherein the shoulder section includes a stress concentration geometry that axially overlaps the strain gauge, the stress concentration geometry being a recess extending circumferentially at least partially around an axis of the bolt; wherein the bolt is configured to orient the strain gauge relative to the weight machine in a predetermined orientation, the bolt including an alignment feature that is a flat face of the shoulder section that mates with a corresponding alignment feature of a bracket of the weight machine, the corresponding alignment feature being a non-circular through-hole; wherein the predetermined orientation includes the strain gauge being in one of tension or shear when the pulley is loaded.
 3. The load sensor of claim 1, wherein the shoulder section includes a stress concentration geometry proximate the strain gauge.
 4. The load sensor of claim 3, wherein the stress concentration geometry axially overlaps the strain gauge.
 5. The load sensor of claim 4, wherein the stress concentration geometry is a recess extending circumferentially at least partially around an axis of the bolt.
 6. The load sensor of claim 1, wherein the bolt is configured to orient the strain gauge relative to the weight machine.
 7. The load sensor of claim 6, wherein the bolt includes an alignment feature the shaft to control orientation of the strain gauge relative to the weight machine, the alignment feature being a flat face of the shoulder section of the bolt.
 8. The load sensor of claim 7, wherein the alignment feature of the bolt mates with a corresponding alignment feature of a bracket of the weight machine, the corresponding alignment feature being a non-circular through-hole that mates with the alignment feature of the bolt to control the orientation.
 9. The load sensor of claim 6, wherein the bolt is configured to orient the load sensor with the strain gauge being in one of tension or shear when the pulley is loaded.
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